

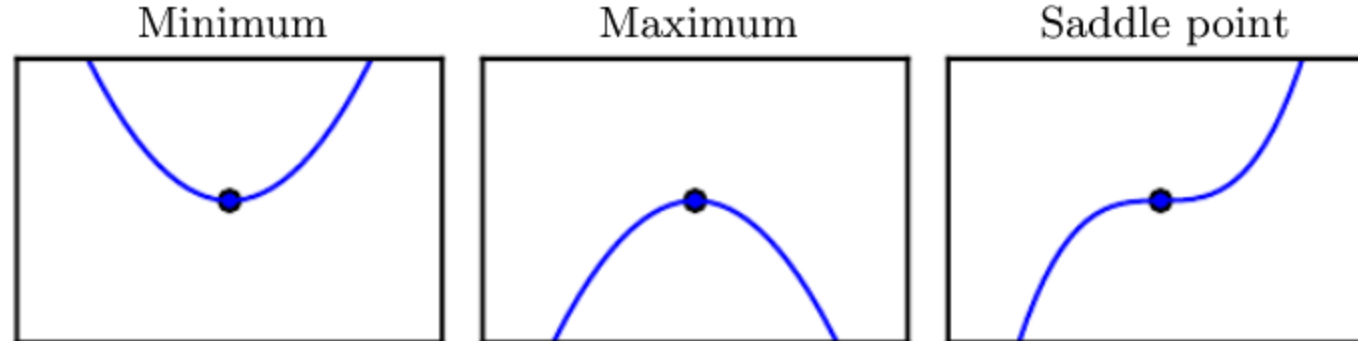
PREREQUISITE REVIEW: CALCULUS

Neural Networks

CALCULUS



- Differential calculus tells us the slope of a function



COMMON RULES FOR DIFFERENTIATION



Power rule:

$$\frac{d}{dx}(x^n) = nx^{n-1}$$

Scalar multiplication rule:

$$\frac{d}{dx}(cu) = c \frac{du}{dx}$$

Sum rule:

$$\frac{d}{dx}(u + v) = \frac{du}{dx} + \frac{dv}{dx}$$

Difference rule:

$$\frac{d}{dx}(u - v) = \frac{du}{dx} - \frac{dv}{dx}$$

COMMON RULES FOR DIFFERENTIATION



Product rule:

$$\frac{d}{dx}(uv) = v \frac{du}{dx} + u \frac{dv}{dx}$$

Quotient rule:

$$\frac{d}{dx}\left(\frac{u}{v}\right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$

Exponential rule:

$$\frac{d}{dx}(e^x) = e^x$$

Logarithm rule:

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

COMMON RULES FOR DIFFERENTIATION



Trigonometry rules:

$$\frac{d}{dx}(\sin x) = \cos x$$

$$\frac{d}{dx}(\cos x) = -\sin x$$

$$\frac{d}{dx}(\tan x) = \sec^2 x$$

$$\frac{d}{dx}(\cot x) = -\csc^2 x$$

$$\frac{d}{dx}(\sec x) = \sec x \tan x$$

$$\frac{d}{dx}(\csc x) = -\csc x \cot x$$

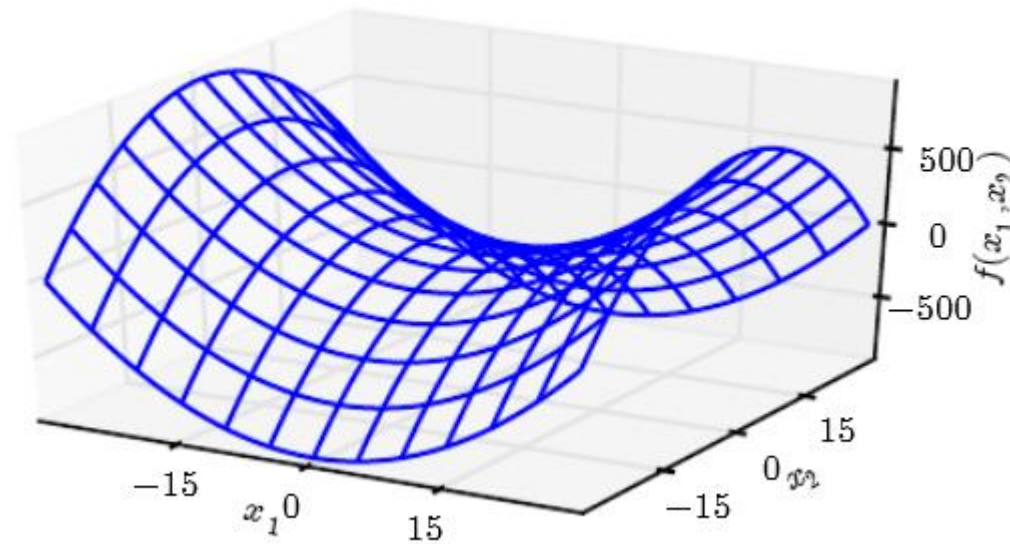
PARTIAL DERIVATIVES

- A measurement of how a multi-variable function changes with respect to one variable

$\frac{\partial f(x,y)}{\partial x}$ differentiates with y held constant

$\frac{\partial f(x_1, \dots, x_n)}{\partial x_i}$ differentiates with all variables except x_i held constant

PARTIAL DERIVATIVES



GRADIENT

- A vector of partial derivatives

$$\nabla f(x, y) = \left[\frac{\partial f(x, y)}{\partial x}, \frac{\partial f(x, y)}{\partial y} \right]$$

$$\nabla f(x_1, \dots, x_n) = \left[\frac{\partial f(x_1, \dots, x_n)}{\partial x_1}, \dots, \frac{\partial f(x_1, \dots, x_n)}{\partial x_n} \right]$$